

DSP Lab Viva — Complete Prep Sheet

Covers your syllabus (DSP intro, DTFT, DFT, properties, convolution, FFT radix-2 DIT/DIF, FIR vs IIR, linear phase, window design) **and** your 7 lab experiments.

How to use tonight: (1) Read Part 1 (formulas) twice. (2) Read Part 2 (theory Q&A). (3) For each experiment in Part 3, be able to say the aim, what the code does line by line, and what your output means. (4) Do the numericals in Part 4 by hand once. (5) Skim Part 5 rapid-fire on the way to the viva.

PART 1 — Formula Cheat Sheet

Topic	Formula
Nyquist rate	$f_s \geq 2 \cdot f_{\max}$
Alias frequency	$f_a = f_m - \text{round}(f_m/f_s) \cdot f_s $
DTFT	$X(e^{j\omega}) = \sum x(n) e^{-j\omega n}, n = -\infty \dots \infty$ (periodic, period 2π)
DFT	$X(k) = \sum x(n) e^{-j2\pi kn/N}, n = 0 \dots N-1$
IDFT	$x(n) = (1/N) \sum X(k) e^{+j2\pi kn/N}$
Twiddle factor	$W_N = e^{-j2\pi/N}; X = W_N \cdot x$
DFT-DTFT link	$X(k) = X(e^{j\omega})$ at $\omega = 2\pi k/N$
Frequency resolution	$\Delta f = f_s/N$
Linear conv length	$L = L_x + L_h - 1$
Circular conv	$y_c(n) = \sum x(k) h((n-k) \bmod N)$
Circ = linear if	$N \geq L_x + L_h - 1$
Time aliasing	$y_c(n) = \sum_m y(n + mN)$
Parseval	$\sum x(n) ^2 = (1/N) \sum X(k) ^2$
Direct DFT cost	N^2 complex mults, $N(N-1)$ complex adds
FFT cost	$(N/2) \cdot \log_2 N$ complex mults, $N \cdot \log_2 N$ complex adds
FIR	$y(n) = \sum h(k) x(n-k), k = 0 \dots M$ ($M+1$ taps)
Linear phase (FIR)	$h(n) = \pm h(M-n) \rightarrow$ group delay = $M/2$ samples
Ideal LPF	$h_d(n) = \sin[\omega_c(n-\alpha)] / [\pi(n-\alpha)], \alpha = M/2$
Windowed FIR	$h(n) = h_d(n) \cdot w(n)$
Hamming	$0.54 - 0.46 \cos(2\pi n/M)$
Hanning	$0.5 - 0.5 \cos(2\pi n/M)$

Topic	Formula
Blackman	$0.42 - 0.5 \cos(2\pi n/M) + 0.08 \cos(4\pi n/M)$
Triangular (Bartlett)	$1 - 2n - M /M$
Bilinear transform	$s = (2/T)(1-z^{-1})/(1+z^{-1}); \Omega = (2/T) \tan(\omega/2)$
Butterworth	$ H(j\Omega) ^2 = 1/[1 + (\Omega/\Omega_c)^{2N}]$
Butterworth order	$N \geq \log[(10^{(0.1R_s)} - 1)/(10^{(0.1R_p)} - 1)] / [2 \log(\Omega_s/\Omega_p)]$
Butterworth cutoff	$\Omega_c = \Omega_p / (10^{(0.1R_p)} - 1)^{1/(2N)}$
Roll-off	20N dB/decade (6N dB/octave)
Hilbert transformer	$H(j\omega) = -j \operatorname{sgn}(\omega): -90^\circ \text{ for } \omega > 0, +90^\circ \text{ for } \omega < 0, H =1$
Analytic signal	$x_a(t) = x(t) + j\hat{x}(t)$
Hilbert pairs	$H\{\cos \omega_0 t\} = \sin \omega_0 t; H\{\sin \omega_0 t\} = -\cos \omega_0 t$

PART 2 — Syllabus Theory with Viva Q&A

2.1 Introduction to DSP

Q1. What is DSP? Processing of signals in digital form: the analog signal is sampled, quantised, processed by a digital processor (DSP chip, FPGA, CPU), and optionally converted back to analog.

Q2. Block diagram of a DSP system? Analog input → **Anti-aliasing filter** → **Sample & hold** → **ADC** → **Digital processor** → **DAC** → **Reconstruction (smoothing) filter** → Analog output.

Q3. Advantages of DSP over analog processing?

- Programmable: change the algorithm by changing software, no hardware change
- Accuracy and repeatability: no component drift, ageing or tolerance problems
- Exactly linear-phase filters are possible (FIR)
- Easy storage of signals, easy to implement adaptive and time-varying systems
- Complex operations (FFT, adaptive filtering, compression) are practical
- Lower cost and higher integration (one chip, many functions)

Q4. Disadvantages / limitations?

- Needs ADC/DAC and anti-aliasing/reconstruction filters
- Limited bandwidth (set by the sampling rate and processor speed)
- Finite word-length effects: quantisation noise, round-off, overflow, limit cycles
- Higher power for very high frequency signals; design complexity

Q5. Applications of DSP? Speech and audio (compression, MP3, noise cancellation), image/video processing, telecommunications (modems, OFDM, software-defined radio), radar/sonar, biomedical (ECG/EEG filtering), seismology, control systems, instrumentation, spectrum analysers, echo cancellation, wireless sensing.

Q6. What is aliasing and how is it prevented? When $f_s < 2f_{max}$, spectral replicas overlap and a high frequency masquerades as a lower one. Prevented by sampling at $f_s \geq 2f_{max}$ and using an analog **anti-aliasing low-pass filter** before the ADC.

Q7. Difference between continuous-time, discrete-time and digital signals? Continuous-time: defined for every t . Discrete-time: defined only at integer n (amplitude continuous). Digital: discrete in both time and amplitude (quantised).

2.2 DTFT (Discrete-Time Fourier Transform)

Q8. Define the DTFT. $X(e^{j\omega}) = \sum x(n) e^{-j\omega n}$, n from $-\infty$ to ∞ . Inverse: $x(n) = (1/2\pi) \int X(e^{j\omega}) e^{j\omega n} d\omega$ over any 2π interval.

Q9. Properties of the DTFT? Continuous in ω , **periodic with period 2π** , generally complex (magnitude even and phase odd for real $x(n)$). Exists if $x(n)$ is absolutely summable.

Q10. Why can't we compute the DTFT directly on a computer? It is a function of continuous ω and needs an infinite sum. So we use the DFT, which samples it at N points and needs only N input samples.

Q11. Relation between DTFT and DFT? The N -point DFT is the DTFT of the finite sequence sampled at $\omega_k = 2\pi k/N$, $k = 0 \dots N-1$.

Q12. DTFT vs DFT: key differences?

DTFT	DFT
Continuous in ω	Discrete in k
Periodic 2π	N -point, implicitly periodic N
Infinite/finite sequences	Finite-length sequences
Not computable directly	Computable (FFT)

2.3 DFT and its Properties

Q13. Define the DFT and IDFT. See Part 1.

Q14. What is the twiddle factor and its properties? $W_N = e^{(-j2\pi/N)}$.

- Periodicity: $W_N^{(k+N)} = W_N^k$
- Symmetry: $W_N^{(k+N/2)} = -W_N^k$
- $W_N^{(N/2)} = -1$, $W_N^N = 1$
- $W_N^2 = W_{N/2}$

Q15. List the DFT properties.

1. **Linearity:** $a \cdot x_1 + b \cdot x_2 \leftrightarrow a \cdot X_1 + b \cdot X_2$
2. **Periodicity:** $X(k+N) = X(k)$, $x(n+N) = x(n)$
3. **Circular time shift:** $x((n-m) \bmod N) \leftrightarrow X(k) \cdot e^{(-j2\pi km/N)}$
4. **Circular frequency shift:** $x(n) \cdot e^{(j2\pi ln/N)} \leftrightarrow X((k-l) \bmod N)$
5. **Circular convolution:** $x_1(n) \circledast x_2(n) \leftrightarrow X_1(k) \cdot X_2(k)$
6. **Multiplication:** $x_1(n) \cdot x_2(n) \leftrightarrow (1/N) X_1(k) \circledast X_2(k)$
7. **Time reversal:** $x((-n) \bmod N) \leftrightarrow X((-k) \bmod N)$
8. **Conjugate symmetry:** for real $x(n)$, $X(N-k) = X^*(k) \rightarrow$ magnitude even, phase odd about $N/2$
9. **Parseval:** $\sum |x|^2 = (1/N) \sum |X|^2$
10. **Duality**

Q16. What does conjugate symmetry tell you in practice? For a real signal only the first $N/2+1$ bins carry independent information. In Exp 3, $|X(1)| = |X(7)| = 10.4525$ and phases are negatives of each other.

Q17. What is $X(0)$? The sum of all samples (DC term). For $\{1 \dots 8\}$, $X(0) = 36$.

Q18. Frequency resolution of the DFT? How to improve it? $\Delta f = f_s/N$. Only a **longer observation window (larger N of actual data)** improves resolution. **Zero padding** gives a denser, interpolated plot of the same spectrum but does not separate closely spaced components.

Q19. What is spectral leakage? When the signal isn't an integer number of periods in the window, energy spreads into neighbouring bins (due to the implicit rectangular window). Reduced by windowing (Hamming, Hanning, etc.).

Q20. Is the DFT periodic? What does that imply? Yes, both $x(n)$ and $X(k)$ are implicitly periodic with N , which is why shifts and convolution become *circular*.

2.4 Linear and Circular Convolution

Q21. What is linear convolution? Output length? $y(n) = \sum x(k)h(n-k)$: the output of an LTI system. Length $L_x + L_h - 1$.

Q22. What is circular convolution? Convolution using modulo- N shifts of two N -point sequences; output also has N points. Equivalent to multiplying DFTs.

Q23. When is circular convolution equal to linear? When $N \geq L_x + L_h - 1$ (both sequences zero-padded to N). Otherwise the tail wraps around (time-domain aliasing).

Q24. Why do we care about this? Fast filtering: pad, FFT, multiply, IFFT gives linear convolution cheaply. For long signals use **overlap-add** or **overlap-save** with FFT blocks.

Q25. Methods to compute circular convolution? (a) Circular-shift (matrix) method, (b) concentric circle method, (c) DFT/IDFT: $y = \text{IDFT}\{X \cdot H\}$, (d) linear convolution then wrap/add the tail, as in your lab.

Q26. Properties of convolution? Commutative, associative, distributive. Convolution with $\delta(n)$ gives the same sequence. Delay in either input delays the output.

2.5 FFT and Radix-2 DIT / DIF

Q27. What is the FFT? Why is it needed? An efficient algorithm to compute the DFT, exploiting symmetry and periodicity of the twiddle factor. Reduces cost from N^2 to $(N/2)\log_2 N$ complex multiplications. Example: $N = 1024 \rightarrow$ direct 1,048,576 vs FFT 5,120 (about 200× faster).

Q28. What is radix-2? N is a power of 2 ($N = 2^v$); the DFT is broken repeatedly into 2-point DFTs. Needs $v = \log_2 N$ stages with $N/2$ butterflies each.

Q29. Explain Decimation-in-Time (DIT) FFT. Split $x(n)$ into **even-indexed** and **odd-indexed** samples: $X(k) = G(k) + W_N^k \cdot H(k)$, $X(k + N/2) = G(k) - W_N^k \cdot H(k)$ where G, H are $N/2$ -point DFTs of the even and odd parts. Repeat until 2-point DFTs.

- **Input:** bit-reversed order; **Output:** natural order
- **Butterfly:** $A = a + W \cdot b$; $B = a - W \cdot b$ (twiddle multiplies *before* the add/subtract)

Q30. Explain Decimation-in-Frequency (DIF) FFT. Split the **output** $X(k)$ into even and odd k . Write the first and second halves of $x(n)$ as $x(n)$ and $x(n+N/2)$: $X(2r) = \text{DFT}_{(N/2)}\{x(n) + x(n+N/2)\}$ $X(2r+1) = \text{DFT}_{(N/2)}\{[x(n) - x(n+N/2)] \cdot W_N^n\}$

- **Input:** natural order; **Output:** bit-reversed
- **Butterfly:** $A = a + b$; $B = (a - b) \cdot W$ (twiddle multiplies *after* the subtract)

Q31. DIT vs DIF?

	DIT	DIF
Splits	Input sequence (time)	Output sequence (frequency)
Input order	Bit-reversed	Natural
Output order	Natural	Bit-reversed
Butterfly	$a \pm W \cdot b$	$(a + b), (a - b) \cdot W$
Complexity	Same: $(N/2)\log_2 N$ mults, $N \cdot \log_2 N$ adds	Same

Q32. For an 8-point FFT: number of stages, butterflies, multiplications, additions? 3 stages, 4 butterflies per stage = 12 butterflies, 12 complex multiplications (fewer if trivial W^0 counted as free), 24 complex additions. Direct DFT: 64 mults.

Q33. Bit reversal for $N = 8$? Index order 0,1,2,3,4,5,6,7 $\rightarrow$ 0,4,2,6,1,5,3,7. (Reverse the 3-bit binary: 001 $\rightarrow$ 100 etc.)

Q34. Twiddle factors for $N = 8$? $W^0 = 1$, $W^1 = (1-j)/\sqrt{2}$, $W^2 = -j$, $W^3 = -(1+j)/\sqrt{2}$, and $W^{(k+4)} = -W^k$.

Q35. What is in-place computation? Each butterfly's two outputs overwrite its two inputs, so only N memory locations are needed.

Q36. How can the FFT compute the IDFT? Conjugate $X(k)$, apply FFT, conjugate result, divide by N . Or swap twiddle sign and scale by $1/N$.

2.6 FIR vs IIR and Linear Phase

Q37. Difference between FIR and IIR?

	FIR	IIR
Impulse response	Finite length	Infinite length
Realisation	Non-recursive (no feedback)	Recursive (feedback)
Transfer function	Only zeros (poles at origin)	Poles and zeros
Stability	Always stable	Can be unstable (poles must be inside unit circle)
Phase	Exactly linear phase possible	Non-linear phase
Order for same spec	High	Much lower
Memory/computation	More	Less
Design	Window, frequency sampling, equiripple	Analog prototype + bilinear / impulse invariance
Quantisation sensitivity	Low	Higher

Q38. What is linear phase and why is it desirable? $\theta(\omega) = -\alpha\omega$, so the group delay $\tau = -d\theta/d\omega = \alpha$ is constant. Every frequency component is delayed equally, so the waveform shape is preserved (**no phase/dispersion distortion**). Important for audio, data communication, image processing, biomedical signals (ECG shape), radar.

Q39. Condition for linear phase in FIR? Symmetric $h(n) = h(M-n)$ (phase = $-\omega M/2$) or **antisymmetric** $h(n) = -h(M-n)$ (phase = $-\omega M/2 + \pi/2$). Group delay = $M/2$ samples.

Q40. The four types of linear-phase FIR?

Type	Symmetry	Length (M+1)	Notes
I	Symmetric	Odd (M even)	All filter types
II	Symmetric	Even (M odd)	Zero at $\omega=\pi \rightarrow$ cannot be high-pass
III	Antisymmetric	Odd	Zero at $\omega=0$ and $\pi \rightarrow$ band-pass, Hilbert, differentiator
IV	Antisymmetric	Even	Zero at $\omega=0 \rightarrow$ high-pass, Hilbert, differentiator

Your lab uses $M = 60$ (61 coefficients, symmetric) = **Type I**, which is why both low-pass and high-pass work.

Q41. Why can't an IIR filter have exactly linear phase? Causal IIR cannot have a symmetric impulse response, since it is infinite and causal. (Zero-phase is possible offline with `filtfilt`.)

Q42. Why are FIR filters always stable? No feedback; $H(z)$ has poles only at $z = 0$, and the output is a finite weighted sum of inputs (BIBO stable).

2.7 FIR Design Using Window Techniques

Q43. Steps of the window method?

1. Choose the ideal response $H_d(\omega)$ and cutoff ω_c (middle of the transition band)
2. Find the ideal impulse response $h_d(n)$ (inverse DTFT) — infinite and non-causal
3. Choose a window and order M that satisfy the transition width and attenuation
4. Truncate to $M+1$ samples and shift by $\alpha = M/2$ for causality: $h(n) = h_d(n) \cdot w(n)$
5. Check the response; adjust M or window if needed

Q44. What is the Gibbs phenomenon? Abrupt truncation (rectangular window) makes the response oscillate near the cutoff with about 9% overshoot; **increasing M does not reduce the peak overshoot**, it only narrows the oscillation region. Smooth windows reduce it.

Q45. Compare the windows.

Window	Main-lobe width	Min stop-band atten.	Transition width
Rectangular	$4\pi/M$	21 dB	$1.8\pi/M$
Triangular	$8\pi/M$	25 dB	$6.1\pi/M$
Hanning	$8\pi/M$	44 dB	$6.2\pi/M$
Hamming	$8\pi/M$	53 dB	$6.6\pi/M$
Blackman	$12\pi/M$	74 dB	$11\pi/M$

Trade-off: narrow main lobe = sharp transition but poor attenuation (rectangular); low side lobes = good attenuation but wide transition (Blackman).

Q46. Why does a window reduce ripple but widen the transition band? Multiplication in time = convolution in frequency. A smooth window has lower side lobes (less ripple) but a wider main lobe (blurs the sharp edge).

Q47. How to design a high-pass FIR? $H_{HP} = \text{all-pass} - H_{LP}$, i.e. $h_{HP}(n) = \delta(n-\alpha) - h_{LP}(n)$. Needs a Type I filter (even M) because a Type II filter has a zero at $\omega = \pi$.

Q48. Disadvantages of the window method? Cannot control ω_p , ω_s and ripples independently; ripples δ_p and δ_s are roughly equal and fixed by the window; not optimal (equiripple/Parks-McClellan is better). The **Kaiser window** gives an adjustable β to trade them off.

Q49. How is the order chosen? From the transition width: $M \approx (\text{window constant})/(\Delta\omega)$. Your lab: transition = 100 Hz = 0.2π rad, $M = 60 \rightarrow$ Blackman needs $11\pi/60 \approx 0.18\pi$, which fits.

Q50. Other FIR design methods? Frequency sampling, Parks-McClellan (equiripple), least-squares.

PART 3 — Experiment-wise Notes and Viva Questions

Experiment 1 — Basic Signals

Aim: generate and plot unit impulse, step, ramp, exponential, rectangular pulse, sine, square, sawtooth, triangular.

Key points

- Time vector `t = (-20000:40000)*1e-4` runs -2 s to 4 s, step 0.1 ms, a fine grid so the curve looks continuous.
- Impulse approximated by a pulse of width $\Delta = 0.05$ s and height $1/\Delta = 20$ (area = 1).
- Step: `double(t>=0)`. Ramp: `t.*(t>=0)`. Exponential: `exp(0.5*t).*(t>=0)` ($a = 0.5 > 0$ so it grows). Rect pulse: `abs(t)<=0.5` so $T = 1$ s.
- Periodic waves at $f_0 = 100$ Hz ($T_0 = 10$ ms), 3 periods = 30 ms. `w = 100*tp` is time in periods.
- Sawtooth: `2*(w - floor(w)) - 1` (ramp from -1 to 1 each period).
- Square: `2*(mod(floor(2*w),2)==0) - 1` (+1 in the first half period, -1 in the second).
- Triangular: `2*abs(saw) - 1`, built from the sawtooth.
- The `1e-9` inside `floor` avoids floating-point errors at period boundaries.

Q&A

- Why can't a true impulse be generated?** Infinite height, zero width; so approximate with a narrow pulse of area 1.
- Relationship between δ , u , r ?** $\delta = du/dt$, $u = \int \delta$, $r = \int u$.
- When does $e^{(at)}$ grow or decay?** Grows for $a > 0$, decays for $a < 0$.
- Period of a 100 Hz wave?** 10 ms.

5. **What is a periodic signal?** $x(t + T_0) = x(t)$ for all t .
 6. **Even/odd?** Rect pulse and triangular (as plotted) are even; sine is odd.
 7. **Energy vs power signal?** Impulse, rect pulse: energy signals; step, ramp (grows), sine, periodic waves: power (ramp/exponential: neither).
 8. **Why `subplot`?** Several plots in one figure window.
 9. **Is the signal you plot truly continuous?** No, it's a densely sampled version.
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Experiment 2 — Sampling Theorem

Aim: verify the sampling theorem at f_s above, at, and below Nyquist; reconstruct; see aliasing.

Key points

- $x(t) = \cos(2\pi \cdot 50t)$, $f_m = 50$ Hz, Nyquist rate = 100 Hz. $f_s = 500, 100, 60$ Hz.
- Reconstruction by **sinc interpolation**: $x_r = x_s * \text{sinc}((t_c - n' \cdot T_s)/T_s)$. MATLAB's $\text{sinc}(x) = \sin(\pi x)/(\pi x)$.
- MSE computed on the middle part (0.05 to 0.15 s) to avoid edge effects of the finite sum.
- Results: $f_s=500 \rightarrow$ MSE 0.00000 (perfect); $f_s=100 \rightarrow$ MSE 0.01674 (small, finite-sum truncation); $f_s=60 \rightarrow$ MSE 1.00849 and apparent frequency **10 Hz** (aliasing).
- **Why cosine, not sine?** A sine at exactly $f_s = 2f_m$ would be sampled at its zero crossings, giving all-zero samples.
- Alias formula: $f_a = |50 - \text{round}(50/60) \cdot 60| = |50 - 60| = 10$ Hz.

Q&A

1. **State the sampling theorem.** A band-limited signal with highest frequency $f_{\max}$ can be exactly recovered if $f_s \geq 2f_{\max}$.
2. **Nyquist rate vs Nyquist frequency?** Rate = $2f_{\max}$; frequency = $f_s/2$.
3. **What does sampling do in the frequency domain?** Replicates the spectrum at every multiple of f_s , scaled by f_s : $X_s(f) = f_s \sum X(f - kf_s)$.
4. **Why does aliasing happen at 60 Hz?** Replicas centred at 0 and 60 Hz overlap; the 50 Hz component folds to $60 - 50 = 10$ Hz.
5. **Why is MSE not exactly zero at $f_s = 100$ Hz?** Ideal reconstruction needs infinite samples; the finite sinc sum has truncation error, and critical sampling is a marginal case (replicas just touch).
6. **Ideal reconstruction filter?** Ideal low-pass with cutoff $f_s/2$; time-domain equivalent is sinc interpolation.
7. **Why can't the ideal reconstruction be built in practice?** Sinc is non-causal and infinite; real DACs use zero-order hold plus an analog smoothing filter.
8. **What is an anti-aliasing filter?** Analog LPF before the ADC removing content above $f_s/2$.

9. **Over-, critical and under-sampling?** $f_s > 2f_m, = 2f_m, < 2f_m$.
 10. **Would 100 Hz or 60 Hz sampling of a 50 Hz sine work?** At 100 Hz the sine starts at a zero crossing and is lost; theoretical requirement is strictly $f_s > 2f_{\max}$ for sinusoids at the edge.
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Experiment 3 — N-point DFT

Aim: compute the N-point DFT and plot magnitude and phase.

Key points

- Code builds the twiddle matrix: $X = x * \exp(-1j*2*\pi/N).^n$. That is $X = W \cdot x$ (W is symmetric so the row-vector form works).
- Input {1,2,3,4,5,6,7,8}: $X(0) = 36$, $X(k) = -4 + j4\cot(\pi k/8)$, e.g. $X(1) = -4 + j9.6569$.
- Compared with `fft`: max difference $6.7e-14$ (floating-point round-off, not a real error).
- Magnitude symmetric ($|X(1)| = |X(7)|$); phase odd. Phase at $k = 4$ prints -180° (the value is $-4 - 0j$; the sign of the tiny negative zero imaginary part decides $\pm 180^\circ$).
- Plots against k (not Hz). Frequency of bin k is $f_k = k \cdot f_s / N$.

Q&A

1. **Define the DFT.** See Part 1.
 2. **Why $N = 8$ and what if the sequence isn't length N ?** Matrix method needs exactly N samples; shorter sequences are zero-padded.
 3. **What is $X(0)$?** Sum of samples = 36.
 4. **Why is $X(4)$ real?** For real $x(n)$ with even N , the Nyquist bin $X(N/2)$ is real: here -4 .
 5. **Why is the magnitude symmetric?** Conjugate symmetry $X(N-k) = X^*(k)$.
 6. **Cost of your program vs `fft`?** Yours is N^2 mults (matrix multiply); FFT is $(N/2)\log_2 N$.
 7. **What does the phase spectrum represent?** Phase of each sinusoidal component; here it reflects the time position of the ramp.
 8. **How is the phase computed?** `angle(X)` = $\text{atan2}(\text{Im}, \text{Re})$, range $(-\pi, \pi]$.
 9. **How do you convert bin k to Hz?** $f = k \cdot f_s / N$.
 10. **What is the DFT of a unit impulse?** All ones (flat spectrum). Of a constant: $N \cdot \delta(k)$.
 11. **What does `stem` do vs `plot`?** `stem` shows discrete samples.
 12. **Why is the difference $6.7e-14$ and not zero?** Different algorithm ordering, finite precision arithmetic.
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Experiment 4 — Linear and Circular Convolution

Aim: perform both and show circular = linear when $N \geq L_x + L_h - 1$.

Key points

- `conv(x,h)` linear; `cconv(x,h,N)` N-point circular.
- Run 1: $x = \{4,5,6,7\}$, $h = \{2,3,4\}$: linear = $\{8, 22, 43, 52, 45, 28\}$ (length 6). 4-point circular = $\{53, 50, 43, 52\}$ because the last two (45, 28) wrap: $8+45 = 53$, $22+28 = 50$.
- Run 2: $x = \{4,3,2,1\}$, $h = \{1,2,3,4\}$: linear = $\{4,11,20,30,20,11,4\}$; 4-point circular = $\{24,22,24,30\}$: $4+20 = 24$, $11+11 = 22$, $20+4 = 24$, 30.
- With $N = L_x + L_h - 1$ (6 and 7), circular equals linear.

Q&A

1. **Why is circular $\neq$ linear for $N = 4$?** $N < L_x + L_h - 1$, so time-domain aliasing occurs.
 2. **Minimum N for equality?** $L_x + L_h - 1$ (6 for run 1, 7 for run 2).
 3. **How can circular convolution be done with DFT?** $y = \text{IDFT}\{\text{DFT}(x) \cdot \text{DFT}(h)\}$, both zero-padded to N.
 4. **Overlap-add vs overlap-save?** Both filter long signals in blocks using FFT: overlap-add adds overlapping output tails; overlap-save discards the corrupted (wrapped) first $M-1$ samples of each block.
 5. **Is linear convolution commutative?** Yes, $xh = hx$.
 6. **Output length of convolving 4 and 3 samples?** 6.
 7. **Sum check:** $\Sigma y = \Sigma x \cdot \Sigma h$ ($22 \times 9 = 198 = 8+22+43+52+45+28$). Good quick sanity check.
 8. **Why does `cconv(x,h,L)` with $L = L_x + L_h - 1$ equal `conv`?** Zero padding provides room so nothing wraps.
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Experiment 5 — FIR Filters (Window Method)

Aim: design FIR LPF and HPF with rectangular, triangular, Hanning, Hamming, Blackman windows; compare ripple and attenuation.

Key points

- $f_s = 1000$ Hz, $M = 60$ (61 taps), $f_p = 200$ Hz, $f_{st} = 300$ Hz.
- Cutoff $f_c = (f_p + f_{st})/2 = 250$ Hz. `$w_n = (f_p + f_{st})/f_s = 0.5$` is normalised to Nyquist ($f_s/2 = 500$ Hz), so $0.5 \times 500 = 250$ Hz.
- `$\text{fir1}(M, w_n, \text{'low'/'high'}, \text{window})$` designs the filter; `$\text{freqz}$` gives the response.
- Pass-band ripple = $\max |\text{gain in dB}|$ for $f \leq f_p$. Stop-band attenuation = $-\max \text{gain}$ for $f \geq f_{st}$.

Window	Ripple (dB)	Stop-band atten. (dB)
Rectangular	0.335	30.05
Triangular	0.199	30.02
Hanning	0.016	54.87
Hamming	0.020	57.99
Blackman	0.002	75.26

- Group delay $M/2 = 30$ samples = 30 ms at $f_s = 1$ kHz.
- **Likely question:** measured attenuation (e.g. rectangular 30 dB) is higher than the table's minimum (21 dB) because the table value is the first side-lobe peak, which falls inside the transition band; you measure from $f_{st} = 300$ Hz onward, past that peak.
Hanning/Hamming/Blackman follow the same pattern.

Q&A

1. **Why is the cutoff in the middle of the transition band?** The windowed response passes through about -6 dB at ω_c and the transition is symmetric about it.
2. **Why M must be even for the high-pass?** Type I filter needed (Type II has a zero at $\omega = \pi$).
3. **Which window is best?** Depends: Blackman for highest attenuation, rectangular for the narrowest transition. Hamming is the common compromise.
4. **Why does the rectangular window have the worst attenuation?** Highest side lobes (-13 dB first lobe), giving about 21 dB minimum attenuation regardless of M .
5. **Does increasing M improve rectangular-window attenuation?** No; it sharpens the transition but the peak ripple/Gibbs stays.
6. **What is fir1's window argument length?** $M+1$.
7. **Why normalise frequency to Nyquist?** MATLAB convention: $1 = f_s/2$.
8. **Transition width in your design?** $100 \text{ Hz} = 0.2\pi \text{ rad}$. Blackman needs $11\pi/60 \approx 0.18\pi$, so it fits.
9. **What's the filter's delay?** 30 samples.
10. **How is the high-pass obtained?** Spectral inversion of the low-pass.
11. **Why are all FIR filters here stable?** No feedback.

Experiment 6 — IIR (Butterworth) Filters

Aim: design Butterworth LPF and HPF for given specs and filter a test sequence.

Key points

- Specs: $R_p = 1$ dB, $R_s = 40$ dB, $f_s = 8000$ Hz. LPF: $f_p = 1000$, $f_{st} = 2000$. HPF: $f_p = 2000$, $f_{st} = 1000$ (note for HPF $f_p > f_{st}$).
- `butterord` gives the minimum order N and cutoff W_n ; `butter` gives coefficients b, a ; `freqz` plots the response; `filter` applies it.
- Results: $N = 6$ for both. LPF cutoff 1105.4 Hz, HPF cutoff 1856.9 Hz. Gains: -1.00 dB at f_p , -40.07 dB at f_{st} (spec met).
- Test signal: tones at $f_s/16 = 500$ Hz and $0.45 \cdot f_s = 3600$ Hz. LPF keeps 500 Hz and removes 3600 Hz; HPF does the opposite.
- Hand check of the order: prewarped $\Omega_p = \tan(\pi \cdot 1000/8000) = 0.414$, $\Omega_s = \tan(\pi \cdot 2000/8000) = 1 \rightarrow$ ratio 2.414. $N \geq \log(9999/0.259) / (2 \cdot \log 2.414) = 4.587/0.765 = 5.99$, so $N = 6$.

Q&A

1. **Butterworth characteristics?** Maximally flat in pass-band, monotonic decrease, no ripple anywhere, -3 dB at cutoff, roll-off $20N$ dB/decade.
2. **Why bilinear transform?** Maps the whole $j\Omega$ axis onto one turn of the unit circle, so **no aliasing**; stable analog filters map to stable digital filters.
3. **Disadvantage of bilinear?** Non-linear frequency warping: $\Omega = (2/T)\tan(\omega/2)$. Hence **pre-warping** of edge frequencies.
4. **Impulse invariance vs bilinear?** Impulse invariance samples $h(t)$: suffers aliasing and only suits band-limited (LPF/BPF) designs; bilinear has no aliasing but warps frequency.
5. **Where are Butterworth poles?** On a semicircle of radius Ω_c in the left-half s -plane, equally spaced.
6. **Stability condition (digital IIR)?** All poles inside the unit circle.
7. **Compare Butterworth, Chebyshev I/II, elliptic?** Butterworth: flat both bands, widest transition. Cheby I: ripple in pass-band. Cheby II: ripple in stop-band. Elliptic: ripple in both, sharpest transition (lowest order).
8. **Why is the order small compared with FIR?** Feedback gives poles which create sharp roll-off cheaply ($N = 6$ vs $M = 60$).
9. **What do `[b, a]` represent?** Numerator and denominator coefficients of $H(z)$.
10. **Why is the HPF's cutoff 1856.9 Hz?** `butterord` places it so the stop-band spec at 1000 Hz and pass-band at 2000 Hz are satisfied with $N = 6$, and `butter` returns the -3 dB point.
11. **LPF to HPF transformation?** Done by `butter(N, Wn, 'high')`, which maps $s \rightarrow \Omega_c/s$ in the analog prototype.
12. **Phase of your filter?** Non-linear; the output sine looks clean here because only one tone remains.

Experiment 7 — Hilbert Transform

Aim: compute the Hilbert transform and analytic signal.

Key points

- $x(t) = \cos(2\pi \cdot 50t)$, $f_s = 1000$ Hz, 1 s, $N = 1000$.
- `hilbert(x)` returns the **analytic signal** (not just the transform); `imag()` of it is the Hilbert transform $= \sin(2\pi \cdot 50t)$. Max error $4.82e-14$.
- Hilbert transformer response built manually: $H(0)=0$, $H(\text{Nyquist})=0$, $-j$ for positive-frequency bins (indices $2 \dots N/2$), $+j$ for negative bins. Phase plot: -90° for $f > 0$, $+90^\circ$ for $f < 0$.
- Ideal discrete Hilbert transformer: $h(n) = (2/\pi n) \sin^2(\pi n/2)$ for $n \neq 0$, $h(0)=0$ (zero for even n).

Q&A

1. **Define the Hilbert transform.** $\hat{x}(t) = x(t) * 1/(\pi t)$; shifts every frequency component by -90° (positive freq) or $+90^\circ$ (negative freq) without changing magnitude.
2. **Frequency response?** $H(j\omega) = -j \operatorname{sgn}(\omega)$.
3. **What is the analytic signal?** $x_a(t) = x(t) + j \hat{x}(t)$; its spectrum is zero for negative frequencies and doubled for positive ones.
4. **Applications?** SSB modulation, quadrature (I/Q) signal generation, envelope and instantaneous frequency detection, demodulation.
5. **Envelope and instantaneous phase?** Envelope $= |x_a(t)|$, phase $= \angle(x_a(t))$, inst. frequency $= (1/2\pi) d(\text{phase})/dt$.
6. **H{cos} and H{sin}?** sin and $-\cos$.
7. **Why H = 0 at DC and Nyquist?** $\operatorname{sgn}(0) = 0$ and a -90° shift of a DC/Nyquist component isn't realisable for real signals.
8. **Is the Hilbert transformer causal?** The ideal one isn't; implement an FIR approximation (Type III/IV antisymmetric) with a delay.
9. **Applying the Hilbert transform twice?** Gives $-x(t)$.
10. **Why `imag(hilbert(x))`?** MATLAB's function returns the full analytic signal.
11. **Why does an FIR Hilbert transformer need antisymmetric $h(n)$?** To get the constant 90° phase shift.

PART 4 — Hand Numericals (Practice Once)

N1. 4-point DFT of $x(n) = \{1, 2, 3, 4\}$ $W_4 = -j$. $X(0) = 10$; $X(1) = 1 + 2(-j) + 3(-1) + 4(j) = -2 + 2j$; $X(2) = 1 - 2 + 3 - 4 = -2$; $X(3) = -2 - 2j$. Answer: $\{10, -2+2j, -2, -2-2j\}$.

N2. Linear convolution of {1,2,3} and {1,1} = {1, 3, 5, 3}. (Length $3+2-1 = 4$.)

N3. 4-point circular convolution of {1,2,3,4} and {1,1,0,0} $y(0) = 1+4 = 5$, $y(1) = 2+1 = 3$, $y(2) = 3+2 = 5$, $y(3) = 4+3 = 7 \rightarrow \{5, 3, 5, 7\}$.

N4. Alias frequency: 70 Hz sampled at 100 Hz $\rightarrow |70 - 1 \cdot 100| = 30$ Hz. 900 Hz sampled at 1000 Hz $\rightarrow 100$ Hz.

N5. FFT counts, $N = 64$: $v = 6$ stages, 32 butterflies per stage, 192 complex mults, 384 complex adds (direct DFT: 4096 mults).

N6. DFT bin spacing: $f_s = 8$ kHz, $N = 1024 \rightarrow \Delta f = 7.8125$ Hz.

N7. Group delay of an FIR with $M = 40$, $f_s = 2$ kHz: $M/2 = 20$ samples = 10 ms.

N8. Sketch the 8-point DIT flow graph: input order $x(0), x(4), x(2), x(6), x(1), x(5), x(3), x(7)$; stage 1 pairs adjacent samples (twiddle W^0); stage 2 uses W^0, W^2 ; stage 3 uses W^0, W^1, W^2, W^3 ; output $X(0) \dots X(7)$ in natural order.

PART 5 — Rapid-Fire (One-Liners)

- **DFT of $\delta(n)$?** 1 for all k .
- **DFT of a constant?** N at $k = 0$, zero elsewhere.
- **FFT requirement?** $N = 2^v$ (radix-2); pad with zeros otherwise.
- **FFT stages for $N = 16$?** 4.
- **Unit of f_s/N ?** Hz (bin spacing).
- **Is circular convolution commutative?** Yes.
- **FIR poles?** At the origin only.
- **Filter that can have exactly linear phase?** FIR.
- **Which is always stable?** FIR.
- **Which needs lower order?** IIR.
- **Best stop-band attenuation among your windows?** Blackman (75 dB).
- **Narrowest main lobe?** Rectangular.
- **Window method disadvantage?** No independent control of edge frequencies and ripples.
- **Hilbert transform of $\cos$?** $\sin$.
- **Bilinear transform drawback?** Frequency warping.
- **Butterworth -3 dB point?** At the cutoff Ω_c , for every order.
- **Roll-off of a 6th-order Butterworth?** 120 dB/decade.

- **MATLAB function names:** `fft`, `ifft`, `conv`, `cconv`, `fir1`, `freqz`, `buttord`, `butter`, `filter`, `hilbert`, `stem`, `subplot`, `fftshift`, `angle`, `abs`.
 - **Abs vs angle?** Magnitude vs phase of complex numbers.
-

PART 6 — Last-Minute Checklist

- ☐ Can I write DFT/IDFT, twiddle properties, and list the DFT properties?
- ☐ Can I draw the 8-point DIT butterfly and state DIT vs DIF differences?
- ☐ Do I know why circular $\neq$ linear and the $N \geq L_x + L_h - 1$ rule?
- ☐ Can I explain linear phase and the four FIR types?
- ☐ Can I compare the five windows (ripple vs transition width)?
- ☐ Can I explain why the measured attenuation in Exp 5 exceeds the table values?
- ☐ Can I explain the bilinear transform and pre-warping?
- ☐ Can I explain each line of the code in my 7 programs without looking?
- ☐ Do I remember my actual output values (Exp 2 MSE, Exp 3 $X(1)$, Exp 4 sequences, Exp 5 table, Exp 6 $N = 6$ and cutoffs)?

Tip: if you don't know an answer, say what you do know about the concept, then connect it to your experiment output. Examiners mostly ask about *your* code and *your* results, so be ready to justify every parameter you chose (f_s , M , W_n , N).

Good luck!